

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

May-June 2024

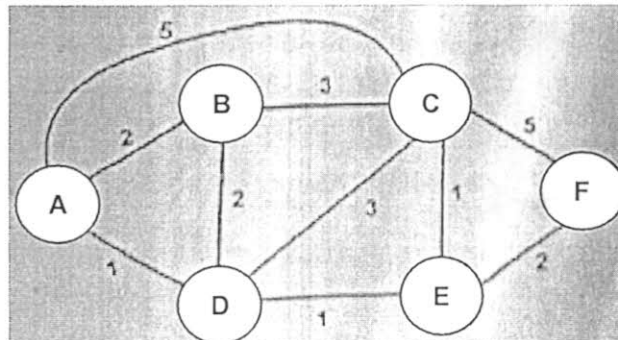
22MCA201 – DATA COMMUNICATION AND NETWORKS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full question** from each unit.

- | | Unit – I | Marks | BT* | CO* | PO* |
|---|-----------------|--------------|------------|------------|------------|
| 1. a) Explain the working of TCP/IP model with a neat diagram. | 10 | L*2 | 1 | 1 | |
| b) Given P=110101 and M=1010001101 find FCS using | | | | | |
| i) modulo 2 arithmetic | 10 | L3 | 1 | 2 | |
| ii) polynomial method | | | | | |
| 2. a) What is modulation? Explain the different digital modulation techniques with examples. | 08 | L2 | 1 | 1 | |
| b) Explain basic communication model with a neat diagram. | 08 | L2 | 1 | 1 | |
| c) Write a note on HDB3 scrambling technique. | 04 | L2 | 1 | 1 | |
| Unit – II | | | | | |
| 3. a) What is ARQ? With diagram distinguish between Go-Back-N-ARQ and Selective Reject ARQ | 10 | L3 | 2 | 1 | |
| b) Explain the frame structure of HDLC. | 10 | L2 | 2 | 1 | |
| 4. a) Explain the sliding window flow control method with an example. | 10 | L2 | 2 | 1 | |
| b) Differentiate between ARM and ABM in HDLC. | 04 | L3 | 2 | 1 | |
| c) What is the use of DNS in application layer, explain the various DNS resource record types for IPv4 | 06 | L2 | 2 | 1 | |
| Unit – III | | | | | |
| 5. a) Differentiate between TCP and UDP. | 08 | L3 | 3 | 1 | |
| b) Explain the steps required to establish and release TCP connections. | 12 | L2 | 3 | 1 | |
| 6. a) Briefly explain the Berkely Socket primitives for TCP. | 10 | L2 | 3 | 1 | |
| b) With neat diagram, explain TCP Segment header and the functions of each field. | 10 | L2 | 3 | 1 | |
| Unit – IV | | | | | |
| 7. a) How are the packets routed using hierarchical routing? Explain. | 08 | L2 | 4 | 1 | |
| b) Apply Dijkstra's routing algorithm to compute the shortest path from node A to all other nodes in the network. | | | | | |



- c) Write a note on optimality principle.

08	L3	4	2
04	L2	4	1

P.T.O

8.	a)	Explain how connectionless services can be implemented in the network.	10	L2	4	1
	b)	Explain the services provided by the network layer to the transport layer.	06	L2	4	1
	c)	Write a note on Store and forward packet switching.	04	L2	4	1

Unit – V

9.	a)	What is Quality of service (QoS)? Discuss the techniques of achieving good quality of service in a network.	12	L2	5	1
	b)	Compare IPV6 header Format with IPV4 Header Format.	08	L3	5	1
10.	a)	Explain different congestion control techniques used in virtual-circuits subnets.	10	L2	5	1
	b)	Differentiate between the leaky bucket and token bucket algorithm. Describe the algorithms with neat diagrams.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
